

Koushik Bala

Kolkata, India

Phone: +91-9679005154 Email: koushikbala54@gmail.com

GitHub: github.com/koushikbala LinkedIn: linkedin.com/in/koushikbalasxc

1 Professional Summary

AI Engineer and Full-Stack Developer with hands-on experience building production-oriented AI learning systems, Retrieval-Augmented Generation (RAG) pipelines, LLM evaluation workflows, responsive web applications, REST APIs, and database-driven platforms. Proficient in Python, FastAPI, React, JavaScript, Node.js, Express.js, MongoDB, Mongoose, Qdrant, embeddings, prompt engineering, and AI/ML integration. Experienced in document ingestion, hybrid retrieval, rubric-based scoring, structured model outputs, grounding, guardrails, role-based portals, cloud deployment, and automated testing.

2 Technical Skills

Programming Languages: Python, JavaScript (ES6+), C, C++, SQL, HTML5, CSS3

Frontend: React, Vite, React Router, Tailwind CSS, Framer Motion, Responsive Design, Accessibility

Backend and APIs: Node.js, Express.js, FastAPI, REST APIs, MongoDB, Mongoose, JWT Authentication, Middleware, Role-Based Access Control

AI Engineering: Python, LLM Integration, Ollama, OpenRouter, Prompt Engineering, AI Orchestration, Structured JSON Outputs, Guardrails, Fallback Design

RAG and Retrieval: Document Ingestion, OCR, Parsing, Chunking, Embeddings, nomic-embed-text, Qdrant, Metadata Filtering, Hybrid Semantic and Keyword Search, Reranking, Citations, Context Grounding

AI Evaluation and ML: Rubric-Based Scoring, Answer Evaluation, Confidence Metrics, Speech and Writing Assessment, PyTorch, Keras, Deep Learning, Data Preprocessing

Speech and Vision AI: Whisper, Faster-Whisper, Speech-to-Text, English, Bengali, Arabic, SpeechBrain, Qwen Vision, Computer Vision, Facial Recognition

Testing and DevOps: Jest, React Testing Library, API Testing, Git, GitHub Enterprise, Docker, Vercel, Linux, CI/CD, AWS Fundamentals, Service Workers, Web Push

Soft Skills: Communication, Teamwork, Adaptability, Time Management, Leadership, Problem-Solving

3 Professional Experience

Core Engineering Team Member and Team Lead

August 2025 – Present

HouseofMusa [CitiMart Group], Kolkata, India

- Built and enhanced an AI-powered Learning Management System (LMS) and ERP platform using React, Tailwind CSS, Express.js, MongoDB, REST APIs, and a Python FastAPI AI service.
- Engineered a RAG pipeline for teacher-approved curriculum material: document parsing and OCR, chunking, embeddings with nomic-embed-text, Qdrant storage, tenant-aware metadata filtering, hybrid retrieval, reranking, and source citations.
- Developed modular AI capabilities for tutor chat, question generation, summaries, flashcards, visual explanations, language practice, and AI-assisted learning workflows through a centralized orchestration layer.
- Implemented AI evaluation workflows for assignment and language submissions using rubric-based scoring, strict structured JSON outputs, measurable confidence signals, and local-model/API fallback handling.
- Built a multilingual speech-to-text and language assessment pipeline for English, Bengali, and Arabic using Whisper/Faster-Whisper, word timestamps, confidence scoring, and pronunciation analysis.
- Developed a Qwen Vision-based facial recognition system for image analysis and AI-powered vision workflows.
- Implemented teacher workflows for assignment creation, student submission review, AI evaluation, grading, feedback, and result publishing.
- Developed role-specific portal navigation and targeted notification flows with module-level counts, visited-state tracking, and parent-module rollups.
- Added automated regression coverage with Jest and React Testing Library, validating user flows, routing, evaluation states, and UI behavior.
- Managed Vercel deployments and GitHub Enterprise version control, improving code delivery speed by 25%.
- Contributed to a scalable online fantasy gaming platform by building responsive interfaces and optimizing API performance.

4 Projects

AI-Powered LMS and ERP Platform

2025–2026

- Built a Python/FastAPI AI service with modular orchestration, RAG-backed tutor chat, curriculum document ingestion, OCR, chunking, embeddings, Qdrant retrieval, prompt grounding, citations, and rubric-based evaluation for teacher and student workflows.

Multilingual Speech-to-Text and Language Assessment

2025–2026

Python, Whisper/Faster-Whisper, SpeechBrain

- Built speech-to-text and language assessment workflows supporting English, Bengali, and Arabic, including word-level timestamps, confidence scores, pronunciation analysis, and structured feedback.

Qwen Vision Facial Recognition System

2025–2026

Python, Qwen Vision, Computer Vision

- Developed a vision-based facial recognition system using a Qwen Vision model for image analysis and AI-assisted identity workflows.

E-commerce Website

2025

React, Express.js, MongoDB

- Developed a full-stack e-commerce platform with secure authentication, payment integration, REST APIs, database persistence, and responsive design.

Deep Learning Model for Rock Classification

2024

Python, PyTorch, Keras

- Built a rock-type classification model with data preprocessing and deep learning techniques for image-based classification.

Zerodha Landing Page Clone

2025

React, Tailwind CSS

- Designed a responsive landing page clone with reusable React components and utility-first styling.

5 Education

Master of Science (M.Sc.) in Computer Science

Expected 2025

St. Xavier's College (Autonomous), Kolkata, India

Bachelor of Science (B.Sc. Hons.) in Computer Science

University of Kalyani, India

6 Certifications and Training

- C++ Basic Certification — CodeChef
- Data Analytics and Blockchain BootCamp — National Institute of Electronics and Information Technology (NIELIT)
- AWS Certified Cloud Practitioner — In Progress

7 Achievements

- Developed and deployed production-ready web applications on cloud platforms.
- Recognized by CitiMart Group for contributions to AI-based ERP development during internship.